

CS 228 : Logic in Computer Science

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Decisions

- ▶ The way forward with Assignments
- ▶ Evaluation Cribs
- ▶ A lite version of the course

FO2Automata : Summary

- ▶ Given a sentence $Q_{x_1} \dots Q_{x_n} \varphi$, replace all $\forall$ in terms of $\exists$ and construct the automaton for the formula $\varphi(x_1, \dots, x_n)$
- ▶ Given the automaton for $\varphi(x_1, \dots, x_n)$, the automaton for $\exists x_i \varphi(x_1, \dots, x_n)$ is obtained by **projecting out** the row of x_i
- ▶ This may result in an NFA
- ▶ Determinize it and complement it to get a DFA for $\neg \exists x_i \varphi(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$
- ▶ Intersect with the regular language where each of $x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n$ are assigned 1 exactly at one position

The Computational Effort

Given NFAs A_1, A_2 each with at most n states,

- ▶ The union has at most $2n$ states
- ▶ Intersection has at most n^2 states
- ▶ The complement has at most 2^n states
- ▶ The projection has at most n states

Cost of determinization : $n + 1$ to 2^n

- ▶ $\Sigma = \{0, 1\}$, languages where the n^{th} bit from the right is a 1.
- ▶ NFA has $n + 1$ states.
- ▶ Corresponding DFA has $\geq 2^n$ states

The Computational Effort

- ▶ $\psi = Q_1 \dots Q_n \varphi$. If $Q_i = \exists$ for all i , then size of A_ψ is same the size of A_φ .
- ▶ When $Q_1 = \exists, Q_2 = \forall, \dots$: each $\forall$ quantifier can create a 2^n blowup in automaton size
- ▶ Size of automaton is

$$2^{2^{2^{2^{2^n}}}}$$

where the tower height k is the quantifier alternation size.

- ▶ This number is indeed a lower bound!

The Automaton-Logic Connection

Given any FO sentence φ , one can construct a DFA A_φ such that $L(\varphi) = L(A_\varphi)$.

Summary

- ▶ Given FO formula φ , build an automaton A_φ preserving the language
- ▶ Satisfiability of FO reduces to non-emptiness of underlying automaton

Satisfiability to Model Checking

- ▶ Satisfiability of FO over words
- ▶ Model checking (what is it?)

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- ▶ Model checking (what is it?)
 - ▶ System abstracted as a model DFA/NFA A
 - ▶ Specification written in FO as formula φ
 - ▶ Does the system model satisfy φ ?
 - ▶ That is, is $L(A) \subseteq L(\varphi)$?
 - ▶ $L(A) \cap \overline{L(\varphi)} = \emptyset$?
- ▶ FO-definable $\subseteq REG$

Next directions in the course

- ▶ Going back to general FO, and discuss the nontermination of the satisfiability checking procedure (Shawn Hedman)
- ▶ Optional topic : Inexpressiveness of FO using EF games (Straubing)
- ▶ MSO logic that can capture exactly regular languages (Wolfgang Thomas AAT)
- ▶ Temporal Logics (only LTL) (Baier-Katoen)
- ▶ Immediate next : MSO

Monadic Second Order Logic (MSO)

Symbols in MSO

Formulae of MSO, over signature τ , are sequences of symbols, where each symbol is one of the following:

- ▶ The symbol $\perp$ called **false**
- ▶ An element of the infinite set $\mathcal{V}_1 = \{x_1, x_2, \dots\}$ of **first order variables**
- ▶ An element of the infinite set $\mathcal{V}_2 = \{X_1, X_2, \dots\}$ of **second order variables** where each variable has arity 1 (**new!**)
- ▶ Constants and relations from τ
- ▶ The connectives $\rightarrow, \wedge, \vee, \neg$
- ▶ The quantifiers $\forall, \exists$
- ▶ Paranthesis

Well formed Formulae

A well-formed formula (wff) over a signature τ is inductively defined as follows:

- ▶ $\perp$ is a wff
- ▶ If t_1, t_2 are terms, then $t_1 = t_2$ is a wff
- ▶ If t_i 's are terms for $1 \leq i \leq k$ and R is a k -ary relation symbol in τ , then $R(t_1, \dots, t_k)$ is a wff
- ▶ If t is either a first order variable or a constant, X is a second order variable, then $X(t)$ is a wff
- ▶ If φ and ψ are wff, then $\varphi \rightarrow \psi, \varphi \wedge \psi, \varphi \vee \psi$ and $\neg\varphi$ are wff
- ▶ If φ is a wff and x is a first order variable, then $(\forall x)\varphi$ and $(\exists x)\varphi$ are wff
- ▶ If φ is a wff and X is a second order variable, then $(\forall X)\varphi$ and $(\exists X)\varphi$ are wff

Free and Bound Variables in MSO

- ▶ Free, Bound Variables and Scope same as in FO
- ▶ In a wff $\varphi = \forall X\psi$ or $\exists X\psi$ every occurrence of X in ψ is bound
- ▶ A sentence is a formula with no free first order and second order variables

Assignments on τ -structures

Assignments

For a τ -structure $\mathcal{A}$, an assignment over $\mathcal{A}$ is a pair of functions (α_1, α_2) , where

- ▶ $\alpha_1 : \mathcal{V}_1 \rightarrow u(\mathcal{A})$ assigns every first order variable $x \in \mathcal{V}_1$ a value $\alpha_1(x) \in u(\mathcal{A})$. If t is a constant symbol c , then $\alpha_1(t)$ is $c^{\mathcal{A}}$.
- ▶ $\alpha_2 : \mathcal{V}_2 \rightarrow 2^{u(\mathcal{A})}$ assigns to every second order variable $X \in \mathcal{V}_2$, $\alpha_2(X) \subseteq u(\mathcal{A})$.

Binding on a Variable

For an assignment $\alpha = (\alpha_1, \alpha_2)$ over $\mathcal{A}$, and $x \in \mathcal{V}_i$, $i = 1, 2$, $\alpha_i[x \mapsto a]$ is the assignment $\alpha_i[x \mapsto a](y) = \begin{cases} \alpha_i(y), & y \neq x, \\ a, & y = x \end{cases}$

Satisfaction

We define the relation $\mathcal{A} \models_{\alpha} \varphi$ (read as φ is true in $\mathcal{A}$ under the assignment α) inductively:

- ▶ $\mathcal{A} \not\models_{\alpha} \perp$
- ▶ $\mathcal{A} \models_{\alpha} t_1 = t_2$ iff $\alpha_1(t_1) = \alpha_1(t_2)$
- ▶ $\mathcal{A} \models_{\alpha} R(t_1, \dots, t_k)$ iff $(\alpha_1(t_1), \dots, \alpha_1(t_k)) \in R^{\mathcal{A}}$
- ▶ $\mathcal{A} \models_{\alpha} X(t)$ iff $\alpha_1(t) \in \alpha_2(X)$ (new)
- ▶ $\mathcal{A} \models_{\alpha} (\varphi \rightarrow \psi)$ iff $\mathcal{A} \not\models_{\alpha} \varphi$ or $\mathcal{A} \models_{\alpha} \psi$
- ▶ $\mathcal{A} \models_{\alpha} (\forall x)\varphi$ iff for every $a \in u(\mathcal{A})$, $\mathcal{A} \models_{\alpha[x \mapsto a]} \varphi$
- ▶ $\mathcal{A} \models_{\alpha} (\forall X)\varphi$ iff for every $S \subseteq u(\mathcal{A})$, $\mathcal{A} \models_{\alpha[X \mapsto S]} \varphi$ (new)

Examples

Recall the signature for the graph structure, $\tau = \{E\}$

- ▶ The graph is 3-colorable

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$$\exists X \exists Y \exists Z (\forall x [X(x) \vee Y(x) \vee Z(x)] \wedge$$

$$\forall x \forall y [E(x, y) \rightarrow \{\neg(X(x) \wedge X(y)) \wedge \neg(Y(x) \wedge Y(y)) \wedge \neg(Z(x) \wedge Z(y))\}])$$

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$$\exists I \{ \forall x \forall y [(\neg(x = y) \wedge I(x) \wedge I(y)) \rightarrow \neg E(x, y)] \wedge$$

$$\exists x_1 \dots x_k [\bigwedge_{i \neq j} \neg(x_i = x_j) \wedge \bigwedge_i I(x_i)] \}$$

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$$\wedge \forall x [(E(x) \vee O(x)) \wedge \neg(E(x) \wedge O(x))]$$

$$\wedge \forall x \forall y [S(x, y) \wedge O(x) \rightarrow E(y)]$$

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MSO on Words : Satisfiability

MSO on Words

- ▶ Signature $\tau = (Q_\Sigma, <, S)$, domain or universe = set of positions of a word
- ▶ MSO over words: Atomic formulae

$$X(x) \mid Q_\Sigma(x) \mid x = y \mid x < y \mid S(x, y)$$

- ▶ Given a MSO sentence φ , $L(\varphi)$ defined as usual
- ▶ A language $L \subseteq \Sigma^*$ is MSO definable iff there is an MSO formula φ such that $L = L(\varphi)$
- ▶ Given an MSO sentence φ , is it satisfiable/valid?

MSO Expressiveness

- ▶ Clearly, $FO \subseteq MSO$
- ▶ $MSO = \text{Regular}$